

RED WING, WI, USA

WMO: 726564

Lat: 44.589N

Lon: 92.485W

Elev: 238

StdP: 98.50

Time Zone: -6.00 (NAC)

Period: 94-19

WBAN: 04967

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -25.0	(c) -22.4	(d) -29.0	(e) 0.3	(f) -23.5	(g) -27.1	(h) 0.3	(i) -21.4	(j) 12.8	(k) -13.6	(l) 11.0	(m) -11.3	(n) 2.6	(o) 300	(p) 0.531

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	11.6	32.7	23.1	31.2	22.4	29.1	21.1	25.3	30.4	24.0	28.7	23.0	27.6	4.3	200

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
23.7	19.1	28.7	22.5	17.7	27.2	21.4	16.5	25.9	78.3	30.8	73.4	29.1	69.1	27.1	30.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) 9.9	(b) 8.5	(c) 7.5	(e) -28.9	(f) 35.6	(g) 3.4	(h) 1.7	(i) -31.3	(j) 36.9	(k) -33.2	(l) 37.9	(m) -35.1	(n) 38.9	(o) -37.6	(p) 40.2		
(a) 9.9	(b) 8.5	(c) 7.5	(e) -29.1	(f) 26.9	(g) 3.3	(h) 1.4	(i) -31.5	(j) 27.9	(k) -33.4	(l) 28.8	(m) -35.3	(n) 29.6	(o) -37.7	(p) 30.6		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	8.0	-9.0	-6.6	0.4	8.0	14.6	20.2	22.7	21.2	17.1	9.8	1.8	-5.6
	DBStd	12.34	7.38	7.07	6.89	5.62	4.95	3.83	3.28	3.24	4.90	5.32	5.96	6.72
	HDD10.0	2284	588	466	307	101	13	0	0	0	5	69	252	484
	HDD18.3	4177	846	699	558	312	138	24	4	9	79	269	496	743
	CDD10.0	1541	0	0	8	42	155	305	394	347	219	64	7	0
	CDD18.3	391	0	0	1	3	21	79	139	98	44	6	0	0
	CDH23.3	3769	0	0	4	38	250	750	1417	865	386	59	0	0
	CDH26.7	1240	0	0	1	9	83	240	527	259	108	14	0	0
Wind	WSAvg	3.0	3.2	3.2	3.3	3.6	3.3	2.8	2.4	2.3	2.7	3.2	3.1	3.0
Precipitation	PrecAvg	864	26	20	50	77	100	131	103	126	101	66	50	30
	PrecMax	1034	59	42	129	150	218	253	247	237	235	168	169	72
	PrecMin	722	7	5	8	21	19	50	34	44	24	22	2	6
	PrecStd	91	15	10	30	39	47	54	49	54	61	41	40	17
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	7.5	12.0	22.3	27.6	32.6	33.8	35.2	33.8	32.4	28.6	20.0	12.1
		MCWB	3.9	8.2	14.8	16.2	20.0	23.2	25.2	23.3	21.8	19.2	13.2	9.8
	2%	DB	4.1	7.5	17.3	22.9	28.8	31.9	32.8	31.3	29.1	24.1	16.4	7.4
		MCWB	1.9	4.7	11.9	13.3	17.7	22.0	24.1	23.0	20.9	16.6	11.3	4.9
	5%	DB	2.5	5.0	13.5	20.9	26.2	29.1	31.3	29.0	27.5	21.4	13.0	5.0
		MCWB	0.9	2.7	9.0	12.0	16.7	20.4	23.2	22.0	19.7	14.8	8.7	2.9
	10%	DB	1.2	2.8	10.0	17.6	23.0	27.5	29.1	27.7	25.9	18.2	11.0	2.8
		MCWB	0.0	1.0	6.8	10.2	15.1	19.7	21.7	20.8	19.0	12.9	7.3	1.3
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	4.5	7.9	16.6	17.9	22.3	25.5	27.6	26.1	24.0	21.9	15.0	9.8
		MCDB	7.2	10.7	20.7	25.5	29.3	31.1	31.9	30.7	29.2	26.3	18.6	12.2
	2%	WB	2.5	5.2	12.8	15.1	19.9	23.8	25.9	24.5	22.4	18.0	12.0	5.1
		MCDB	4.3	7.4	16.5	21.6	25.8	28.7	31.4	29.0	27.2	22.5	15.3	7.1
	5%	WB	1.0	3.0	9.6	13.0	18.5	22.5	24.6	23.4	21.2	15.9	9.5	3.1
		MCDB	2.3	4.8	13.3	18.5	23.6	27.3	29.4	27.5	25.5	19.8	12.9	4.6
	10%	WB	-0.1	1.3	6.6	11.2	17.0	21.2	23.3	22.3	19.9	13.8	7.5	1.6
		MCDB	1.0	3.0	9.5	16.7	21.5	26.0	28.0	26.5	24.4	18.3	10.1	3.0
Mean Daily Temperature Range	5% DB	MDBR	9.1	9.7	9.9	11.6	11.9	11.3	11.6	11.2	11.8	10.9	8.9	8.0
		MCDBR	9.7	9.8	14.0	16.5	15.4	13.6	12.7	12.4	13.3	15.0	12.8	9.7
		MCWBR	8.0	7.6	9.0	8.8	7.0	6.0	5.8	5.8	6.6	8.0	8.4	7.8
	5% WB	MCDBR	9.0	9.1	12.9	13.6	12.6	11.3	11.2	10.8	11.1	11.9	11.5	8.9
		MCWBR	7.8	7.3	8.9	8.4	6.7	6.2	6.0	6.1	6.5	7.6	8.1	7.4
Clear-Sky Solar Irradiance	taub	0.261	0.295	0.323	0.373	0.406	0.404	0.397	0.403	0.372	0.325	0.289	0.266	
	taud	2.409	2.304	2.363	2.298	2.252	2.298	2.335	2.327	2.384	2.501	2.507	2.436	
	Ebn at Noon	874	892	912	888	865	866	868	849	848	852	834	829	
	Edh at Noon	72	97	106	123	133	128	122	119	103	80	66	64	
All-Sky Solar Radiation	RadAvg	1.73	2.81	3.81	4.57	5.31	5.92	6.26	5.24	4.11	2.63	1.68	1.22	
	RadStd	0.19	0.31	0.34	0.49	0.42	0.47	0.21	0.34	0.28	0.35	0.16	0.12	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (52 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page